



REPORT

ON

COMPUTER NETWORK

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Subject : Computer Network

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Date: 25 May

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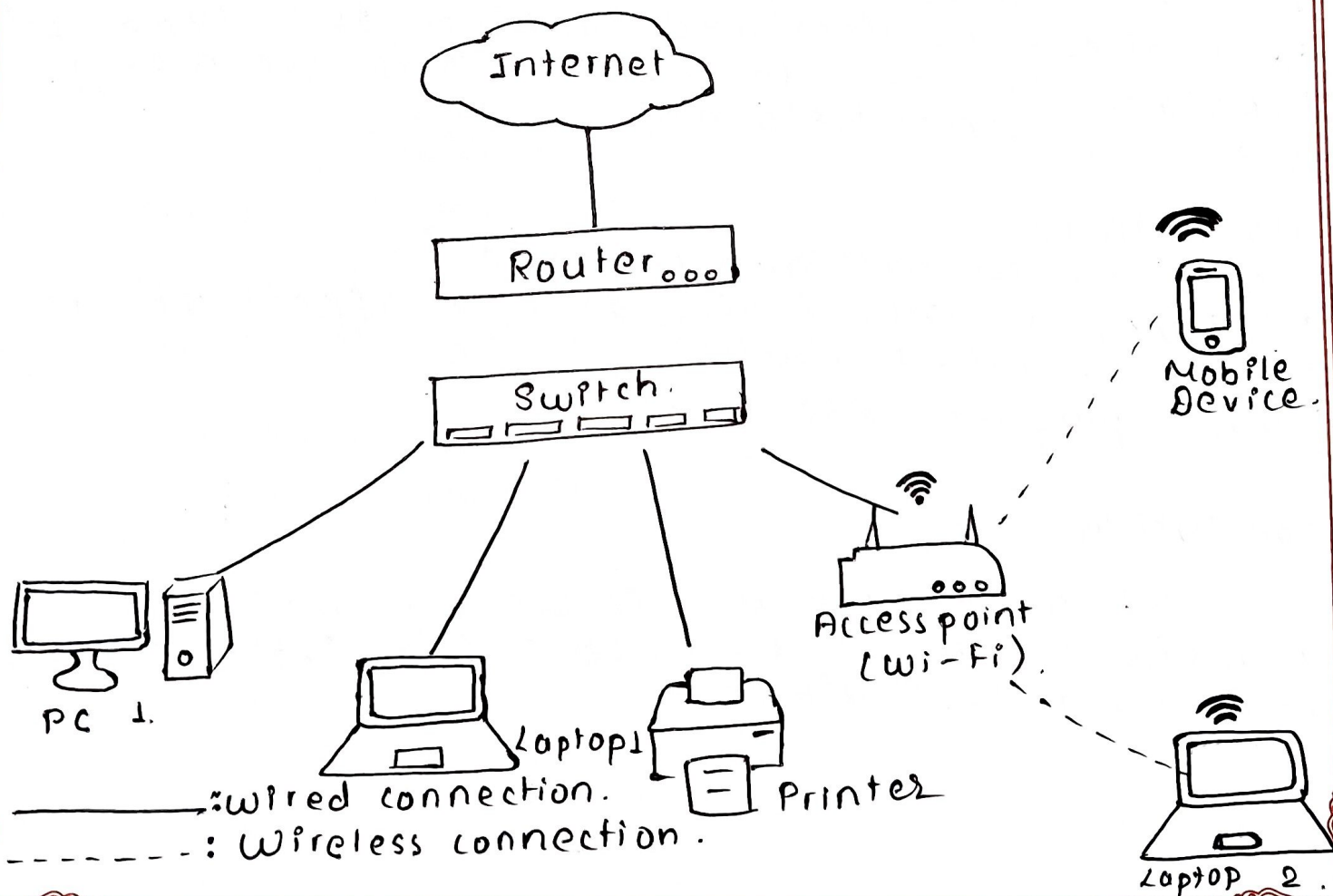
1. Draw the Network Topology:

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- Identify and draw the LAN topology of your organization (e.g. Star, Bus, Mesh).
- Clearly label all components such as Router, Switch, Access Point and end devices (PCs, laptops, etc).

Network Topology (Star Topology):

Aim: To study and draw the LAN network topology used in our organization.



Components used:

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- Router
- Switch
- Access point (Wi-Fi)
- PC
- Laptop.
- Printer
- Mobile Devices.

The network topology used is Star Topology, all devices are connected to a central device such as a switch or router. The router connects the network to the internet, and the switch connects multiple devices using wired connections. An access point is used to provide wireless connectivity to mobile devices and laptops. This type of topology is easy to manage and commonly used in LAN networks.

Advantages.

- Easy to install and manage.
- Failure of one device does not affect others.
- Easy to detect faults.
- Good performance.

Conclusion:

The star topology is widely used in modern networks because it is reliable, efficient and easy to maintain.

2. Identification of Network Devices in LAN:

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1. Introduction:

A local Area Network (LAN) consists of multiple interconnected devices that enable communication, data sharing, and resource utilization within a limited geographical area such as a home, office, or campus. Various networking devices are used to establish, manage and maintain this communication efficiently.

2. List of Networking Devices in LAN:

2.1 Router:

Function:

- Connect different networks (e.g. LAN to the Internet)
- Routes data packets based on IP address.
- Provides internet access to multiple devices.
- Often includes built-in firewall and DHCP services.

2.2 Switch:

Function:

- Connects multiple devices within a LAN.
- Transfers data using MAC addresses.
- Reduces network collisions by sending data only to the intended device.
- Improves network efficiency.

2.3 Hub

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Function:

- connects multiple devices in a network (basic device)
- Broadcasts data to all connected devices.
- works on physical layer (layer 1).
- less efficient compared to devices switches.

2.4 Bridge

Function:

- Connects two LAN segments.
- Filters traffic based on MAC addresses.
- Reduces unnecessary data transmission.
- Works at Data Link Layer (Layer 2).

2.5 Repeater

Function:

- Regenerates and amplifies weak signals.
- Extends the transmission distance of a network.
- Works at Physical Layer.

2.6 Access Point

Function:

- Provides wireless connectivity to devices.
- Connects wireless devices to a wired LAN.
- Commonly used in Wi-Fi networks.

2.7 Modem

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Functions:

- Converts digital signals to analog and vice versa.
- Connects a network to an Internet Service Provider (ISP).
- Required for internet access in many setups.

2.8 Network Interface Card (NIC):

Functions:

- Allows a device to connect to a network.
- Provides a unique MAC address.
- Can be wired (Ethernet) or wireless (Wi-Fi).

2.9 Firewall (Hardware / Software)

Functions:

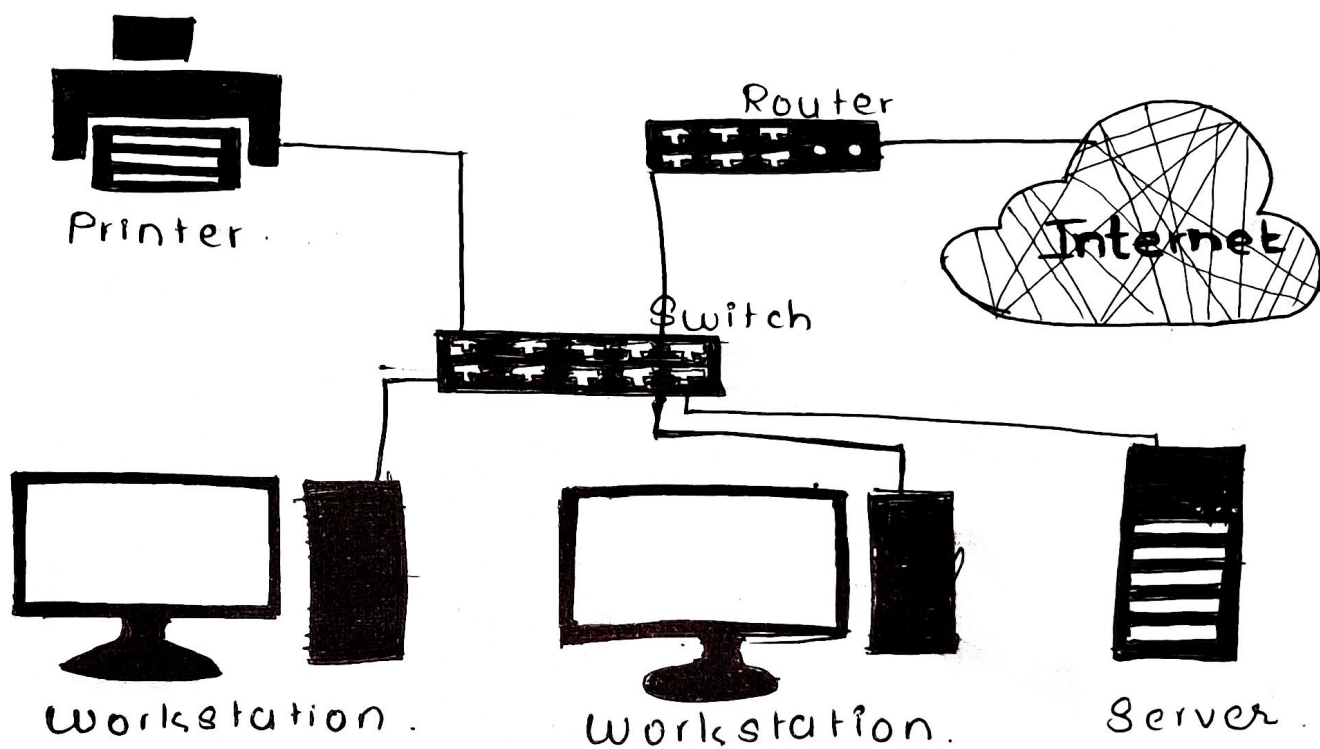
- Protect the network from unauthorized access.
- Monitors incoming and outgoing traffic.
- Enhances network security.

CONCLUSION:

Networking devices play a crucial role in the functioning of a LAN. Each device has a specific purpose, from connecting devices and routing data to ensuring security and efficient communication. Proper understanding and configuration of these devices are essential for maintaining a reliable and secure network.

LAN Network Diagram:

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3. Analyze Connection Types:

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Specify whether connections are:

- Wired (Ethernet)
- Wireless (Wi-Fi)

How to analyze connection type?

~~Way~~
~~Step~~ 1: Check Physically

✓ If Ethernet cable is connected.

- You will see a LAN cable (RJ45) connected to your device.
- Connection type = Wired (Ethernet).

✓ If No cable, using Wi-Fi.

- No cable connected.
- You see Wi-Fi icon on screen.
- Connection type = Wireless (Wi-Fi).

Way 2: Check in your laptop (Windows).

1) Go to settings.

2) Click Network & Internet.

3) See : If it shows Ethernet → Wired.
If it shows Wi-Fi → Wireless

Way 3: Using CMD.

1) Open command prompt.

- Press Windows + R
- Type cmd & press enter

Step 2: Type this command

ipconfig or netsh interface
show interface

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Press enter.

Step 3: Analyze the output.

Analysis of my connection type (using Command Prompt).

Method used:

The Command Prompt was used to analyze the network connection type using the ipconfig command. ~~command~~ netsh interface show interface command.

Observations:

Admin state	State	Type	Interface name
Enabled	Disconnected	Dedicated	Ethernet
Enabled	connected	Dedicated	Ethernet 3
Enabled	connected	Dedicated	Wi-Fi

The output displayed is that both "Ethernet 3" and "Wi-Fi" interfaces were in a connected state, while the default Ethernet interface was disconnected. The active network connection is wireless (Wi-Fi).

Result:

The system is primarily connected using a wireless (Wi-Fi) connection, as the Ethernet interface is active. Physical cable is connected.

Conclusion:

The wireless connection provides better flexibility & higher speed compared to wired connection making

suitable for reliable network usage.
Reference photo:

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Microsoft Windows [Version 10.0.26200.8457]
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C:\Users\swati>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . : D3-WDS03.COM

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::7a20:ea00:c08a:bcdd%4
IPv4 Address. : 192.168.56.1
Subnet Mask : 255.255.255.0
Default Gateway :

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv4 Address. : 192.168.1.66
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.254

C:\Users\swati>netsh interface show interface

Admin State	State	Type	Interface Name
Enabled	Disconnected	Dedicated	Ethernet
Enabled	Connected	Dedicated	Ethernet 3
Enabled	Connected	Dedicated	Wi-Fi

- Mention where each type is used.

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In a local Area Network (LAN), devices can be connected using either wired or wireless communication methods. Each type of connection has its own advantages, limitations and specific use cases.

Types of connections

1) Wired connection (Ethernet)

A wired connection uses physical cables (Ethernet cable) to connect devices within a network.

Features:

- High speed and stable connection.
- Low interfaces.
- More secure compared to wireless.

Where it is used?

- Offices and companies.
- Computer labs.
- Servers and data centers.
- Gaming setups requiring stable internet.

2) Wireless connection (Wi-Fi)

A wireless connection uses radio waves to connect devices without physical cables.

Features:

- Easy to install and use.
- Provides mobility.
- Can connect multiple devices easily.

Where it is used?

- Homes and Hostels.
- Cafes and public places.
- Mobile devices (Phones, tablets, laptops)
- Smart devices (IoT)

4. Measure Network Speed:

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- Find the link speed of:
 - wired network (if available)
 - wireless network

4.1 Objective

To measure the link speed of the wireless (Wi-Fi) network using command Prompt.

4.2 Methods used.

The command Prompt was used to obtain detailed information about the wireless network, including the transmission (T_x) and reception (R_x) link speed.

4.3 Command used.

netsh wlan show interface.

4.4 Observation.

After executing the command, the following details were observed.

- Name: Wi-Fi
- State: Connected
- SSID: dressmebutique_fpkhr
- Receive rate (Mbps): 300
- Transmit rate (Mbps): 300.

4.5 Result

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is

- Receive speed = 300 Mbps
- Transmit speed = 300 Mbps.

These values represent the data transfer rate between the system and wireless router.

4.6 Conclusion

The system is connected using a wireless network. The Command Prompt was used to measure the wireless network speed using the command `netsh wlan show interfaces`. The receive rate was 300 Mbps and the transmit rate was 300 Mbps. This indicates a stable wireless connection suitable for normal usage.

- Mention the tools/commands used (e.g. Task manager, netsh; netsh wlan show interfaces or, etc).

To analyze the network connection type and measure the network speed, various system tools and command-line utilities were used. These tools help in obtaining accurate and detailed information about the network.

Tools used:

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1) Command Prompt (CMD).

Purpose:

- To analyze network interfaces
- To check connection type (wired/wireless).
- To measure network speed.

2) Task Manager:

Purpose

- To monitor real-time network usage.
- To observe data transfer speed.

Commands Used:

1. ipconfig

Function:

- Display IP configuration details.
- Helps identify whether connection is ethernet or Wi-Fi.

2. netsh interface show interface.

Function:

- Shows all network interfaces and their status.
- Helps determine which connection is active.

3. netsh wlan show interfaces.

Function:

Displays detailed information about the wireless network.
Shows Receive and Transmit speed (Mbps).

5. Find System Network Details:

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- Determine and record the following of your computer:

- MAC address
- IP Address (IPv4)

- Mention the commands used (eg ipconfig, getmac)

5.1 Objective:

To determine and record the system's network details such as MAC address, IPv4 address and subnet Mask using command line tools.

5.2 Method used:

The command prompt was used to retrieve network configuration details using specific commands.

5.3 Commands Used.

The command used are:

getmac — for mac address.

ipconfig — for ip address.

5.4 Observation.

After executing the above commands, the following details were obtained:

- MAC Address: 50-13B-135-7B-10-D2
- IPv4 Address: 192.168.1.66
- Subnet Mask: 255.255.255.0

Conclusion:

The system network details were successfully obtained using command prompt. These details are essential for identifying and communicating with devices in a network.

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Reference photo:

```
Microsoft Windows [Version 10.0.26200.8457]
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C:\Users\swati>getmac

Physical Address      Transport Name
=====
50-BB-B5-7B-10-D2    \Device\Tcpip_{BF96F734-34A8-44E1-A662-0533DEB91247}
BC-FC-E7-D2-81-9C    Media disconnected
0A-00-27-00-00-04    \Device\Tcpip_{1EF7B74B-4144-4986-969F-FFCA9ED94A2A}

C:\Users\swati>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : D3-WDS03.COM

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . : fe80::7a20:ea00:c08a:bcdd%4
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : worldlink.com.np
    IPv4 Address. . . . . : 192.168.1.66
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%20
                                192.168.1.254
```

6. Explain Observations:

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• Briefly explain:

- Difference between ~~wired~~ wired and wireless speed.
- Role of subnet mask in networking

Difference between wired and wireless speed.

Wired and wireless networks differ significantly in terms of speed and performance.

• Wired connection (Ethernet):

- Provides higher and more stable speed.
- Less interference and signal loss.
- Suitable for tasks requiring high bandwidth.

• Wireless connection (Wi-Fi):

- Provides comparatively lower speed.
- Speed may vary due to distance and
- Suitable for mobile devices and general usage.

Conclusion:

Wired connections are faster and more reliable, while wireless connections offer flexibility and ease of use.

Role of Subnet Mask in Networking

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The Subnet mask plays an important role in network communication:

- It divides an IP address into network part and host part.
- Helps in identifying whether a device belongs to the same network or not.
- Enables efficient routing of data within a network.
- Reduces network congestion by organizing devices into subnets.

Example:

If the IP address is 192.168.1.5 and subnet mask is 255.255.255.0,

- Network portion $\rightarrow$ 192.168.1.
- Host portion $\rightarrow$ 5.

Conclusion:

The analysis shows that while wired networks provide better speed and stability, wireless networks are more convenient. Additionally, subnet masks are crucial for efficient data transmission and network management.